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Lab 05 – Data Visualization, EDA and Preprocessing

Objectives

- Create line, bar, scatter, histogram, box, and heatmap plots with Matplotlib and Seaborn
- Follow the steps of Exploratory Data Analysis (EDA)
- Detect and handle missing values and outliers
- Encode categorical data and scale numeric data
- Produce a complete **EDA report** on a real dataset

Tools required: Google Colab (Matplotlib, Seaborn, Pandas, and scikit-learn are already installed)

Part 1: Matplotlib Basics

Matplotlib draws charts. Always import it like this:

```
import matplotlib.pyplot as plt
```

Line plot

```
days = [1, 2, 3, 4, 5]
temp = [30, 32, 31, 35, 34]

plt.plot(days, temp, marker="o")
plt.title("Temperature over 5 Days")
plt.xlabel("Day")
plt.ylabel("Temperature (C)")
plt.grid(True)
plt.show()
```

Bar chart

```
subjects = ["Math", "Physics", "English", "CS"]
marks = [72, 65, 80, 91]

plt.bar(subjects, marks, color="steelblue")
plt.title("Marks by Subject")
plt.ylabel("Marks")
plt.show()
```

Scatter plot (relationship between two numbers)

```
hours = [1, 2, 3, 4, 5, 6, 7, 8]
score = [35, 45, 50, 58, 66, 72, 80, 88]

plt.scatter(hours, score)
plt.title("Study Hours vs Score")
plt.xlabel("Hours studied")
plt.ylabel("Score")
plt.show()
```

Histogram (distribution of one numeric column)

```
import numpy as np

ages = np.random.normal(30, 8, 500)      # 500 random ages around 30

plt.hist(ages, bins=20, edgecolor="black")
plt.title("Age Distribution")
plt.xlabel("Age")
plt.ylabel("Number of people")
plt.show()
```

Which plot should I use?

Question	Plot
How does a value change over time?	Line plot
How do categories compare?	Bar chart
Are two numeric columns related?	Scatter plot
How is one numeric column spread out?	Histogram or box plot
Which columns are related to each other?	Heatmap

Task 1.1

1. Draw a line plot of the square of the numbers 1 to 10, with a title and axis labels
2. Draw a bar chart of the marks of 5 students
3. Draw a scatter plot of two lists of your own choice

Part 2: Seaborn Plots

Seaborn is built on Matplotlib and makes good-looking statistical charts with less code. It also includes sample datasets.

```
import seaborn as sns
import matplotlib.pyplot as plt
```

```
df = sns.load_dataset("penguins")
print(df.head())
```
The penguins dataset has the columns `species`, `island`, `bill_length_mm`, `bill_depth_mm`, `flipper_length_mm`, `body_mass_g`, and `sex`.

Count plot (how many of each category)

```
sns.countplot(data=df, x="species")
plt.title("Penguins per Species")
plt.show()
```

Histogram with a smooth curve

```
sns.histplot(data=df, x="body_mass_g", kde=True)
plt.show()
```

Box plot (spread and outliers)

The box shows the middle 50% of the values. Dots outside the whiskers are possible outliers.

```
sns.boxplot(data=df, x="species", y="body_mass_g")
plt.show()
```

Scatter plot with colours

```
sns.scatterplot(data=df, x="flipper_length_mm", y="body_mass_g", hue="species")
plt.show()
```

Correlation heatmap

Correlation measures how strongly two numeric columns move together. It ranges from -1 to +1. A value near +1 means they rise together, and a value near -1 means one rises while the other falls.

```
corr = df.select_dtypes(include="number").corr()
sns.heatmap(corr, annot=True, cmap="coolwarm")
plt.title("Correlation Heatmap")
plt.show()
```

Pair plot (all relationships at once)

```
sns.pairplot(df, hue="species")
plt.show()
```

Task 2.1

Load the dataset `sns.load_dataset("tips")` and draw: a histogram of `total_bill`, a box plot of `total_bill` by `day`, and a scatter plot of `total_bill` against `tip`.

Part 3: Exploratory Data Analysis (EDA)

EDA means getting to know your data **before** building any model. Follow these steps in order:

1. **Look at the data:** `head()`, `shape`, `info()`, `describe()`

2. **Check missing values:** `isnull().sum()`
3. **Check duplicates:** `duplicated().sum()`
4. **Study each column alone:** histograms and count plots
5. **Study relationships:** scatter plots, box plots, heatmap
6. **Find outliers:** box plots or the IQR rule
7. **Write your findings** in plain sentences

```
import pandas as pd
import seaborn as sns

df = sns.load_dataset("penguins")

print(df.shape)
df.info()
print(df.describe())
print(df.isnull().sum())
print("Duplicates:", df.duplicated().sum())
print(df["species"].value_counts())
```

Task 3.1

Run the code above. Write down: how many rows and columns the data has, which columns have missing values, and which species is the most common.

Part 4: Preprocessing

Machine learning models need clean numeric data. Preprocessing prepares the data.

4.1 Missing values

```
df = sns.load_dataset("penguins")

# Numeric columns: fill with the median
num_cols = ["bill_length_mm", "bill_depth_mm", "flipper_length_mm", "body_mass_g"]
for col in num_cols:
    df[col] = df[col].fillna(df[col].median())

# Text column: fill with the most common value
df["sex"] = df["sex"].fillna(df["sex"].mode()[0])

print(df.isnull().sum())
```

4.2 Outliers (IQR rule)

A value is an outlier if it is far outside the middle range of the data.

```
col = "body_mass_g"
q1 = df[col].quantile(0.25)
q3 = df[col].quantile(0.75)
iqr = q3 - q1

lower = q1 - 1.5 * iqr
```

```
upper = q3 + 1.5 * iqr

outliers = df[(df[col] < lower) | (df[col] > upper)]
print("Number of outliers:", len(outliers))

# Option: keep only the values inside the range
df_no_outliers = df[(df[col] >= lower) & (df[col] <= upper)]
```
Careful: Do not remove an outlier automatically. First ask whether it is a mistake or a real, rare value.

4.3 Encoding categorical columns

Models cannot use text such as "Male" or "Adelie". Convert it to numbers.

```
# Two categories: map to 0 and 1
df["sex_num"] = df["sex"].map({"Male": 0, "Female": 1})

# Many categories: one-hot encoding (one 0/1 column per category)
encoded = pd.get_dummies(df, columns=["island"], dtype=int)
print(encoded.head())
```

4.4 Feature scaling

Columns with large numbers (like body mass in grams) can dominate columns with small numbers. Scaling puts the columns on a similar range.

```
from sklearn.preprocessing import StandardScaler, MinMaxScaler

cols = ["bill_length_mm", "body_mass_g"]

# StandardScaler: mean 0, standard deviation 1
std = StandardScaler()
scaled_std = std.fit_transform(df[cols])

# MinMaxScaler: values between 0 and 1
mm = MinMaxScaler()
scaled_mm = mm.fit_transform(df[cols])

print(pd.DataFrame(scaled_std, columns=cols).describe())
print(pd.DataFrame(scaled_mm, columns=cols).describe())
```

Preprocessing step	Why it is needed
Fill missing values	Models cannot use empty cells
Handle outliers	Extreme values can distort results
Encode categories	Models need numbers, not text
Scale features	Keeps large-valued columns from dominating

Task 4.1

1. Count how many outliers `flipper_length_mm` has using the IQR rule
2. One-hot encode the `species` column and print the first 5 rows

3. Scale `flipper_length_mm` with `MinMaxScaler` and confirm that its minimum is 0 and its maximum is 1

Part 5: Lab Project – Complete EDA Report

Goal: Produce an EDA report on the `penguins` dataset. Use one notebook with a text cell (a heading and a short explanation) before every code cell.

Report structure

1. Introduction. What the dataset is about and what each column means.

2. Data overview.

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

df = sns.load_dataset("penguins")
print(df.shape)
print(df.dtypes)
print(df.describe())
```

3. Data quality.

```
print(df.isnull().sum())
print("Duplicates:", df.duplicated().sum())
```

4. Single-column analysis.

```
fig, axes = plt.subplots(1, 3, figsize=(15, 4))
sns.countplot(data=df, x="species", ax=axes[0])
sns.histplot(data=df, x="body_mass_g", kde=True, ax=axes[1])
sns.countplot(data=df, x="sex", ax=axes[2])
plt.show()
```

5. Relationships.

```
sns.scatterplot(data=df, x="bill_length_mm", y="bill_depth_mm", hue="species")
plt.show()

sns.boxplot(data=df, x="species", y="flipper_length_mm")
plt.show()

sns.heatmap(df.select_dtypes(include="number").corr(), annot=True, cmap="coolwarm")
plt.show()
```

6. Cleaning. Fill the missing values (as in Part 4.1) and check for outliers.

7. Preprocessing. Encode `sex` and `island`, and scale the numeric columns.

8. Conclusions. Write at least 5 sentences. For example: which columns separate the species best, which columns are strongly correlated, and what problems you found in the data.

Common Errors (Cheat Sheet)

Error	Cause	Fix
Plot does not appear	Forgot `plt.show()`	Add `plt.show()` after the plot
`ValueError: could not convert string to float`	A text column was passed to a numeric function such as `corr()`	Use `df.select_dtypes(include="number")`
Heatmap shows no numbers	`annot=True` is missing	Add `annot=True`
`KeyError` in `map()` result becomes `NaN`	Mapping keys do not match the data exactly (for example `"male"` vs `"Male"`)	Print `df["sex"].unique()` first
Scaled data has different results each time	Scaler was fit on different data	Fit the scaler once, then reuse it

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Lab Tasks for Student Practice

Solve the following 5 tasks on your own. Paste your code and a screenshot of the output for each task in your lab file.

Task 1: Sales Charts (Easy)

Using a list of 6 months and a list of sales figures, draw a line plot and a bar chart of the same data. Add titles and axis labels to both.

Task 2: Tips Dataset Explorer (Easy)

Load `sns.load_dataset("tips")`. Draw a count plot of `day`, a histogram of `tip`, and a box plot of `total_bill` for lunch and dinner (`time`). Write one sentence about each chart.

Task 3: Correlation Study (Medium)

On the `tips` dataset, draw a correlation heatmap. Which two numeric columns have the strongest correlation? Draw a scatter plot of that pair and describe the relationship.

Task 4: Clean and Prepare (Medium)

Load `sns.load_dataset("titanic")`. Then:

1. Fill the missing `age` values with the median and `embarked` with the mode
2. Drop the `deck` column
3. Encode `sex` as 0/1 and one-hot encode `embarked`
4. Scale `age` and `fare` using `StandardScaler`

Task 5: Full EDA Report (Challenging)

Choose the `diamonds` dataset (`sns.load_dataset("diamonds")`) or any CSV dataset of your choice. Write a complete EDA report using the 8-section structure from Part 5. Include at least 6 different charts and a conclusions section.

Submission Instructions

- Paste the **code and output** of every task in your lab file
- Write **comments** in your code wherever needed
- Name your notebook: `Lab5_RollNumber.ipynb`
- Complete **Tasks 1 to 5** before the next lab

End of Lab 05. Next lab: Introduction to Machine Learning and Regression